

Dependency-Colored Concentration for Finite-Alphabet Shared-Exclusions PID

Finite-sample law control, drift separation, local atom moduli,
and a formal evidence boundary

pid-rs project analysis

23 July 2026

Abstract

This document gives project-defined validation, new in `pid-rs`, for the categorical shared-exclusions partial information decomposition (SxPID) defined by Makkeh, Gutknecht, and Wibral. The paper-defined functional is not new. The document derives a finite-sample concentration bound for the complete empirical joint law when all complete rows share one common finite law and a fixed coloring partitions observations into classes whose complete rows are mutually independent. Dependence across classes can be arbitrary. A class-size proxy that is optimal within the declared Hölder–Hoeffding proof scheme improves the ordinary color-count factor for unbalanced classes. A telescoping error allocation gives a time-uniform envelope. A nonidentical-law extension separates random error from explicit drift. A stronger common-support modulus then transfers joint-law error to pointwise and averaged SxPID atoms. The local result includes one- Λ cumulative control, general-source Möbius-row bounds, and sharper complete two-source atom bounds. Lean checks deterministic algebraic subclaims, and an executable challenge suite retains counterexamples to weaker assumptions. The result does not validate continuous shared exclusions, arbitrary time series, adaptive evaluation transforms, or binary64 asymptotics.

Contents

1	Status, scope, and provenance	2
2	Finite setting and the dependence contract	3
3	Finite-sample law concentration	3
4	Time-uniform envelope and consistency	5
5	Nonidentical row laws and explicit drift	6
6	Strengthened common-support local SxPID continuity	7
7	Composition with SxPID	11
8	Counterexamples and invalidated routes	12
9	Formal and executable evidence boundary	14
10	Ecosystem use boundary	15
11	Reproduction	15
12	Conclusion	15

1 Status, scope, and provenance

The mathematical functional, the concentration ingredients, and this repository artifact have different origins.

Object	Origin	Status in this document
Categorical SxPID	Paper-defined by Makkeh, Gutknecht, and Wibral; its lattice has a formal-logic basis due to Gutknecht, Wibral, and Makkeh [1, 2]	Implemented for two to four sources; receives a finite-sample bound and an almost-sure exact-real consistency implication when the displayed envelope vanishes
Hoeffding bounded-variable inequality	Paper-defined classical result [3]	Used inside each independent color class
Dependency-color concentration	Published background includes Janson’s coloring method and later Hölder formulations [4, 5]	Project-defined proof composition with a class-size proxy that is optimal within the declared Hölder–Hoeffding proof scheme; no novelty claim
Empirical-law L^1 deviation	Published independent and identically distributed (i.i.d.) background includes Weissman et al. [6]	One-sided subset union bound under the stated color contract
SxPID local modulus	Project-defined validation of the paper-defined functional	One- Λ cumulative control, general-source Möbius-row bounds, and sharper complete two-source atom-specific bounds. Sharpness evidence is limited to the stated endpoints, the asymptotic coefficient, and one bounded near-tightness fixture case; none proves global sharpness
Drift envelope	Project-defined extension	Concentration about the average row law plus explicit bias to a reference law

The machine-readable authority is `method-catalog.json`. The rendered authority is `METHODS.md`. The Rust implementation routes covered by the final composition are the stable categorical functions

`discrete_sxpid2`, `discrete_sxpid2_averaged`, `discrete_sxpid3`,
`discrete_sxpid3_averaged`, `discrete_sxpid_n`, and `discrete_sxpid_n_averaged`.

They are exported from `stable::categorical`. This document adds no estimator and no public Rust API. Its proof and executable artifacts are validation evidence.

2 Finite setting and the dependence contract

Let $Z_1, \dots, Z_n$ take values in a finite alphabet $\mathcal{Z}$ with $|\mathcal{Z}| = K$. For SxPID, one row is the complete joint symbol

$$Z_i = (S_{1i}, \dots, S_{mi}, T_i). \quad (1)$$

Dependence between source and target coordinates inside a row is unrestricted. It is the signal that PID analyzes.

For the common-law theorem, assume every row has marginal law P . Let a deterministic color map partition the indices into occupied classes $C_1, \dots, C_r$. Define

$$n_j = |C_j| > 0, \quad n = \sum_{j=1}^r n_j. \quad (2)$$

The complete rows $\{Z_i : i \in C_j\}$ must be mutually independent for every j . Dependence across different classes can be arbitrary. The color map must not depend on the observed row values.

Define

$$\hat{P}_n(z) = \frac{1}{n} \sum_{i=1}^n \mathbf{1}\{Z_i = z\}, \quad V_n = \left(\sum_{j=1}^r \sqrt{n_j} \right)^2, \quad d_{\text{eff},n} = \frac{V_n}{n}. \quad (3)$$

Proposition 2.1 (Effective-color range). *The deterministic proxy obeys*

$$n \leq V_n \leq rn, \quad 1 \leq d_{\text{eff},n} \leq r. \quad (4)$$

Proof. For the lower bound, expand the square of a sum of nonnegative square roots and retain its diagonal terms. For the upper bound, apply Cauchy–Schwarz:

$$\left(\sum_j \sqrt{n_j} \right)^2 \leq \left(\sum_j 1^2 \right) \left(\sum_j n_j \right) = rn. \quad (5)$$

The Lean artifact proves both inequalities over exact real numbers. $\square$

The value $d_{\text{eff},n}$ is a proof constant derived from the declared class sizes. It is not an estimated effective sample size.

3 Finite-sample law concentration

Theorem 3.1 (Dependency-colored empirical-law bound). *Assume $K \geq 2$ and the contract in Section 2. Then, for every $\varepsilon > 0$,*

$$\mathbb{P}\left(\left\|\hat{P}_n - P\right\|_1 \geq \varepsilon\right) \leq \min\left\{1, (2^K - 2) \exp\left(-\frac{n^2 \varepsilon^2}{2V_n}\right)\right\}. \quad (6)$$

If at most d colors are occupied, then

$$\mathbb{P}\left(\left\|\hat{P}_n - P\right\|_1 \geq \varepsilon\right) \leq \min\left\{1, (2^K - 2) \exp\left(-\frac{n \varepsilon^2}{2d}\right)\right\}. \quad (7)$$

Proof. Fix a nonempty proper subset $A \subset \mathcal{Z}$. Put

$$Y_i = \mathbf{1}\{Z_i \in A\} - P(A), \quad S_j = \sum_{i \in C_j} Y_i. \quad (8)$$

Let $R = \sum_j \sqrt{n_j}$ and select the Hölder exponents

$$p_j = \frac{R}{\sqrt{n_j}}, \quad \sum_j \frac{1}{p_j} = 1. \quad (9)$$

Each $p_j \geq 1$ because $R \geq \sqrt{n_j}$, so these are valid Hölder exponents. For $\lambda > 0$, generalized Hölder gives

$$\mathbb{E} \exp \left(\lambda \sum_j S_j \right) \leq \prod_j \left(\mathbb{E} e^{p_j \lambda S_j} \right)^{1/p_j}. \quad (10)$$

Rows are mutually independent inside each color. Hoeffding's lemma for a centered random variable with range length one gives

$$\mathbb{E} e^{p_j \lambda S_j} \leq \exp \left(\frac{n_j p_j^2 \lambda^2}{8} \right). \quad (11)$$

Therefore,

$$\mathbb{E} \exp \left(\lambda \sum_j S_j \right) \leq \exp \left(\frac{\lambda^2 V_n}{8} \right). \quad (12)$$

Chernoff's method, optimized at $\lambda = 4nt/V_n$, yields

$$\mathbb{P} \left(\hat{P}_n(A) - P(A) \geq t \right) \leq \exp \left(-\frac{2n^2 t^2}{V_n} \right). \quad (13)$$

For any signed finite measure with total mass zero,

$$\frac{1}{2} \left\| \hat{P}_n - P \right\|_1 = \max_{A \subseteq \mathbb{Z}} \{ \hat{P}_n(A) - P(A) \}. \quad (14)$$

Set $t = \varepsilon/2$ and take a union bound over the $2^K - 2$ nonempty proper subsets. This proves Equation (6). Complement subsets already encode the opposite sign, so no additional factor of two is required. Proposition 2.1 gives Equation (7). $\square$

For $K = 1$, the L^1 distance is identically zero, and the finite-sample formulas reduce to the true but empty statement $0 \leq 0$ for every $\varepsilon > 0$. The time-uniform radius contains $\log(2^K - 2)$, so Section 4 requires $K \geq 2$. The subset factor is exponential in K , so the bound can be vacuous on a large alphabet. This result makes no sharpness or sample-complexity claim. If $s = |\text{supp}(P)| \geq 2$ is known before the data are inspected, then $2^s - 2$ can replace $2^K - 2$. An observed support size is not a known population support size. When $s = 1$, the law distance is zero and the result is vacuous.

Proposition 3.2 (Hölder proxy optimum). *For $q_j > 0$ with $\sum_j q_j^{-1} = 1$,*

$$V_n \leq \sum_j n_j q_j. \quad (15)$$

Equality holds for $q_j = R/\sqrt{n_j}$.

Proof. The exponent conditions force $q_j \geq 1$ for every j . Thus, they are valid Hölder exponents. Cauchy–Schwarz gives

$$\left(\sum_j \sqrt{n_j} \right)^2 = \left(\sum_j \sqrt{n_j q_j} \sqrt{q_j^{-1}} \right)^2 \leq \left(\sum_j n_j q_j \right) \left(\sum_j q_j^{-1} \right). \quad (16)$$

The selected exponents make the two Cauchy–Schwarz vectors proportional. $\square$

This is an optimum inside the Hölder–Hoeffding proof scheme for this partition. It is not a claim that V_n is the best possible constant for every admissible joint law.

4 Time-uniform envelope and consistency

Assume $K \geq 2$. For an infinite sequence, require one fixed map from the positive integers to a finite or countable color set. Only finitely many colors can be occupied in each prefix. The entire infinite collection in each color must consist of mutually independent complete rows. Let V_n use the nonempty class sizes in the first n rows. A uniform bound of at most d occupied colors is required only for the coarse radius in Equation (23).

Fix $0 < \alpha < 1$ and define

$$C_K = 2^K - 2, \quad \alpha_n = \frac{\alpha}{n(n+1)}. \quad (17)$$

The allocation has the exact finite identity

$$\sum_{n=1}^N \alpha_n = \alpha \left(1 - \frac{1}{N+1}\right) \leq \alpha. \quad (18)$$

Lean proves the unit-scale finite identity by induction. Multiplication by the fixed scalar α gives Equation (18).

Define

$$R_n = \sqrt{\frac{2V_n}{n^2} \log\left(\frac{C_K n(n+1)}{\alpha}\right)}, \quad (19)$$

$$D_n = \min\{2, R_n\}. \quad (20)$$

Theorem 4.1 (Anytime law envelope). *Under the infinite-sequence contract,*

$$\mathbb{P}(\forall n \geq 1 : \|\hat{P}_n - P\|_1 \leq D_n) \geq 1 - \alpha. \quad (21)$$

Proof. For every n , substitution in Theorem 3.1 gives

$$C_K \exp\left(-\frac{n^2 R_n^2}{2V_n}\right) = \alpha_n. \quad (22)$$

The Lean artifact checks the algebraic exponent cancellation. Positivity, the square-root substitution, and its probability interpretation remain in this prose proof. A union bound and Equation (18) give simultaneous control by R_n . Every probability-law L^1 distance is also at most two. Thus it is at most $D_n = \min\{2, R_n\}$. $\square$

The coarser d -color radius is

$$D_n^{(d)} = \min\left\{2, \sqrt{\frac{2d}{n} \log\left(\frac{C_K n(n+1)}{\alpha}\right)}\right\}. \quad (23)$$

Under one uniform fixed bound d , this is $O(\sqrt{d \log(n)/n})$. For each positive rational error threshold, the tails in Equation (7) are summable. Borel–Cantelli and a countable intersection over those thresholds therefore give

$$\|\hat{P}_n - P\|_1 \longrightarrow 0 \quad \text{almost surely.} \quad (24)$$

If the number of colors grows, the displayed envelope converges to zero under the sufficient condition

$$\frac{V_n \log n}{n^2} \longrightarrow 0. \quad (25)$$

This condition is not necessary for empirical-law convergence. For example, i.i.d. rows still converge when a needlessly fine singleton coloring gives $V_n = n^2$. The condition is sufficient for this declared-color envelope. A singleton coloring also permits complete dependence across rows, so the theorem cannot infer convergence from that coloring alone.

Corollary 4.2 (Fixed-width overlapping windows). *Let E_t be i.i.d. innovations. Fix $w \in \mathbb{N}$ before evaluation, and let*

$$Z_t = f(E_t, \dots, E_{t+w-1}) \quad (26)$$

for one fixed measurable finite-output map f . Coloring row t by $t \bmod w$ satisfies the dependence contract with at most $d = w$ colors.

Proof. Two rows in the same color use disjoint innovation blocks. Every finite collection in that color is therefore mutually independent. The fixed map and the i.i.d. innovation law give the common row law. $\square$

This corollary covers a fixed categorical overlapping-window construction from i.i.d. innovations. It does not cover circular wraparound windows, a width selected from evaluation outcomes, a rolling-start sequence of repeated claims, or innovations with unspecified time dependence.

Definition 4.3 (Strong ℓ -dependence). Let $\ell \in \mathbb{N}_0$. An ordered sequence is strongly ℓ -dependent when

$$\sigma(Z_i : i \leq t) \text{ is independent of } \sigma(Z_i : i \geq t + \ell + 1) \quad (27)$$

for every t .

Coloring index t by $t \bmod (\ell + 1)$ separates consecutive members of each color by more than ℓ . Repeatedly separate the last row from the sigma-field generated by all preceding rows. This ordered induction proves mutual independence of all complete rows in each color. Pairwise lag independence alone does not prove this property. The common-law theorem still requires one common row law. If row laws differ, the drift theorem and its additional premises apply.

5 Nonidentical row laws and explicit drift

Assume $K \geq 2$. Let row i have law P_i , and keep the within-color mutual-independence contract. Define

$$\bar{P}_n = \frac{1}{n} \sum_{i=1}^n P_i. \quad (28)$$

Center the subset indicator for row i by $P_i(A)$. The Hölder–Hoeffding proof is unchanged.

Theorem 5.1 (Average-law and reference-law bounds). *For every $\varepsilon > 0$,*

$$\mathbb{P}\left(\left\|\hat{P}_n - \bar{P}_n\right\|_1 \geq \varepsilon\right) \leq \min\left\{1, C_K \exp\left(-\frac{n^2 \varepsilon^2}{2V_n}\right)\right\}. \quad (29)$$

For an explicit reference law $P_\star$, put

$$b_n = \left\|\bar{P}_n - P_\star\right\|_1. \quad (30)$$

Then

$$\mathbb{P}\left(\left\|\hat{P}_n - P_\star\right\|_1 \geq b_n + \varepsilon\right) \leq \min\left\{1, C_K \exp\left(-\frac{n^2 \varepsilon^2}{2V_n}\right)\right\}. \quad (31)$$

Proof. Equation (29) follows from the row-specific centering. The triangle inequality gives

$$\left\|\hat{P}_n - P_\star\right\|_1 \leq \left\|\hat{P}_n - \bar{P}_n\right\|_1 + b_n, \quad (32)$$

which proves Equation (31). $\square$

If $\|P_i - P_\star\|_1 \leq \beta_i$, then convexity gives

$$b_n \leq \frac{1}{n} \sum_{i=1}^n \beta_i. \quad (33)$$

Under one fixed infinite row-law sequence and coloring with the same premises, the all-prefix reference-law envelope is

$$\mathbb{P}(\forall n \geq 1 : \|\hat{P}_n - P_\star\|_1 \leq \min\{2, b_n + D_n\}) \geq 1 - \alpha. \quad (34)$$

The bias term is necessary. Concentration about $\bar{P}_n$ does not identify a fixed scientific estimand.

6 Strengthened common-support local SxPID continuity

This section records claim SXPID-TV-001, revision 2. It gives project-defined validation of a paper-defined functional. It does not define a new PID measure or estimator. All statements use exact real arithmetic.

Let p and q be laws on one fixed complete joint alphabet. Define

$$\delta = \|q - p\|_1, \quad \eta = \frac{\delta}{2}, \quad p_{\min} = \min_{z \in \text{supp}(p)} p(z), \quad L = \log(1/p_{\min}). \quad (35)$$

Assume

$$\text{supp}(q) \subseteq \text{supp}(p), \quad 0 \leq \delta < 2p_{\min}. \quad (36)$$

The convention $\eta = \sup_E |q(E) - p(E)|$ is total variation. Equation (36) implies equal support. Put

$$p_s = (1 - s)p + sq, \quad \mu_s = p_{\min} - s\eta, \quad 0 \leq s \leq 1. \quad (37)$$

Every supported cell under p_s has mass at least $\mu_s > 0$.

Lemma 6.1 (Path oscillation). *Let $\Delta = q - p$. For every real-valued function g on the finite common support,*

$$\left| \sum_z \Delta(z) g(z) \right| \leq \eta \left(\max_z g(z) - \min_z g(z) \right). \quad (38)$$

If a differentiable functional F satisfies $\text{osc } \nabla F(p_s) \leq 1/\mu_s$, then

$$|F(q) - F(p)| \leq \int_0^1 \frac{\eta}{p_{\min} - s\eta} ds = \log \frac{p_{\min}}{p_{\min} - \eta} =: \Lambda(\delta, p_{\min}). \quad (39)$$

Proof. Subtract the midpoint of the range of g in Equation (38) and use $\sum_z \Delta(z) = 0$. Apply that inequality to the directional derivative of F along p_s . The strict support margin makes the complete segment positive, so the fundamental theorem of calculus and direct integration give Equation (39). $\square$

Fix a supported realization $z = (s_1, \dots, s_m, t)$ and a nonempty antichain α of nonempty source subsets. Define

$$A_\alpha = \bigcup_{a \in \alpha} \{S_a = s_a\}, \quad B_\alpha = A_\alpha \cap \{T = t\}, \quad C = \{T = t\}. \quad (40)$$

By definition, $B_\alpha = A_\alpha \cap C$, and the keyed realization belongs to B_α . For a law Q , define

$$c_\alpha^+(z; Q) = -\log Q(A_\alpha), \quad (41)$$

$$c_\alpha^-(z; Q) = \log \frac{Q(C)}{Q(B_\alpha)}, \quad (42)$$

$$c_\alpha^{\text{sx}}(z; Q) = \log \frac{Q(B_\alpha)}{Q(A_\alpha)Q(C)}. \quad (43)$$

Proposition 6.2 (One- Λ cumulative modulus). *For $u \in \{+, -, \text{sx}\}$,*

$$|c_\alpha^u(z; q) - c_\alpha^u(z; p)| \leq \Lambda. \quad (44)$$

Proof. Write $A = A_\alpha$ and $B = B_\alpha$. The cell-gradient values are

Region	∇c^+	∇c^-	∇c^{sx}
B	$-1/Q(A)$	$1/Q(C) - 1/Q(B)$	$1/Q(B) - 1/Q(A) - 1/Q(C)$
$A \setminus C$	$-1/Q(A)$	0	$-1/Q(A)$
$C \setminus A$	0	$1/Q(C)$	$-1/Q(C)$
outside $A \cup C$	0	0	0

The three gradient diameters are at most $1/Q(B)$. For the net column, compare every pair and use $Q(B) \leq Q(A), Q(C)$. Since $p_s(B) \geq \mu_s$, Lemma 6.1 proves the result. The direct net calculation avoids a nonoptimal factor of two. $\square$

Let M be the fixed Möbius matrix of the complete redundancy lattice. Define

$$\pi_\alpha^u(z; Q) = \sum_\beta M_{\alpha\beta} c_\beta^u(z; Q), \quad r_\alpha = \sum_\beta |M_{\alpha\beta}|, \quad (45)$$

$$\bar{\pi}_\alpha^u(Q) = \sum_{z \in \text{supp}(Q)} Q(z) \pi_\alpha^u(z; Q). \quad (46)$$

Also define

$$h = \log \frac{2}{1 + p_{\min}}, \quad J = L - 2h \geq 0. \quad (47)$$

Theorem 6.3 (General-source atom bounds). *For every complete-lattice node and $u \in \{+, -, \text{sx}\}$,*

$$|\pi_\alpha^u(z; q) - \pi_\alpha^u(z; p)| \leq r_\alpha \Lambda. \quad (48)$$

The averaged component bounds are

$$|\bar{\pi}_\alpha^\pm(q) - \bar{\pi}_\alpha^\pm(p)| \leq \min \{L, r_\alpha \Lambda + \eta L\}. \quad (49)$$

The averaged net bound is

$$|\bar{\pi}_\alpha^{\text{sx}}(q) - \bar{\pi}_\alpha^{\text{sx}}(p)| \leq \min \{2L, r_\alpha \Lambda + \delta \min \{L, r_\alpha (L - h)\}\}. \quad (50)$$

Proof. Equation (48) follows from Proposition 6.2 and the triangle inequality. Makkeh, Gutknecht, and Wibral prove that every informative and misinformative pointwise atom is nonnegative on the complete finite antichain lattice [1, Theorem IV.3 and Appendix A]. Their base-2 sign result is unchanged after conversion to nats. Each component atom is therefore at most its top-node component sum, which is at most L . Hence, under p ,

$$0 \leq \pi_\alpha^+, \pi_\alpha^- \leq L, \quad -L \leq \pi_\alpha^{\text{sx}} \leq L. \quad (51)$$

For a cumulative net value, write $a = p(A_\alpha)$, $b = p(B_\alpha)$, and $c = p(C)$. Then $a, c \geq b \geq p_{\min}$ and $a + c - b \leq 1$. Thus

$$ac \leq \frac{(a+c)^2}{4} \leq \frac{(1+b)^2}{4}, \quad (52)$$

and

$$-J \leq \log \frac{b}{ac} \leq L. \quad (53)$$

The cumulative range width is $2(L - h)$. A Möbius row gives net-atom oscillation at most $2r_\alpha(L - h)$; Equation (51) also gives oscillation at most $2L$.

Split an average change into a pointwise-value term and a probability-weight term. Center the second term as in Equation (38). The component weight term is at most ηL . The net weight term is at most $\delta \min\{L, r_\alpha(L - h)\}$. This proves the inner bounds. For the outer caps, the common support has at most $1/p_{\min}$ cells. Each averaged component atom is nonnegative and at most its top component sum, which is a source entropy or conditional source entropy. It is therefore at most L under either law. Each averaged net atom lies in $[-L, L]$. $\square$

The lattice's sole bottom Möbius row has $r_\alpha = 1$, so its net weight coefficient is $L - h$. Every nonbottom row is a nonzero integer vector with zero sum. Thus it has at least one positive and one negative entry, so $r_\alpha \geq 2$. Since $J = L - 2h \geq 0$, its inner net coefficient simplifies to L .

Theorem 6.4 (Complete two-source improvement). *For every complete two-source atom and $u \in \{+, -, \text{sx}\}$,*

$$|\pi_\alpha^u(z; q) - \pi_\alpha^u(z; p)| \leq \Lambda. \quad (54)$$

The averaged bounds are

Atom	Component average change	Net average change
Redundancy	$\min\{L, \Lambda + \eta L\}$	$\min\{2L, \Lambda + \delta(L - h)\}$
Either unique atom	$\min\{L, \Lambda + \eta L\}$	$\min\{2L, \Lambda + \delta(L - h)\}$
Synergy	$\min\{L, \Lambda + \eta J\}$	$\min\{2L, \Lambda + \delta J\}$

Proof. Let $E_i = \{S_i = s_i\}$, $U = E_1 \cup E_2$, and $C = \{T = t\}$. Redundancy uses one event log, one nested log ratio, or one intersection-PMI expression, so Proposition 6.2 applies.

For one unique atom, put

$$N(a, b) = \log \frac{a + b}{a}. \quad (55)$$

Its two nonzero gradient values are

$$D_a = \frac{1}{a + b} - \frac{1}{a}, \quad D_b = \frac{1}{a + b}, \quad (56)$$

whose diameter is $1/a$. For a path law $Q = p_s$, put

$$x_a = Q(E_i \cap C), \quad x_b = Q((U \setminus E_i) \cap C), \quad (57)$$

$$y_a = Q(E_i \cap C^c), \quad y_b = Q((U \setminus E_i) \cap C^c). \quad (58)$$

The keyed cell gives $x_a \geq \mu_s > 0$. Addition is componentwise. Define

$$G_N(x, y) = N(x + y) - N(x). \quad (59)$$

For $i \in \{a, b\}$, the y_i -gradient coordinate is $F_i = D_i(x + y)$ and the x_i -gradient coordinate is $X_i = F_i - D_i(x)$. A coordinate outside the two event regions has gradient value zero.

The $F_a - F_b$ difference is at most $1/(x_a + y_a) \leq 1/x_a$. The same-index differences $F_i - X_i = D_i(x)$ have magnitude at most $1/x_a$. Also,

$$X_a - X_b = \frac{1}{x_a} - \frac{1}{x_a + y_a}, \quad (60)$$

and the two crossed differences are

$$F_a - X_b = -\frac{1}{x_a + y_a} + D_b(x), \quad F_b - X_a = \frac{1}{x_a + y_a} + D_a(x). \quad (61)$$

Each crossed difference is a difference of two values in $[0, 1/x_a]$. Each of F_a, F_b, X_a, X_b has magnitude at most $1/x_a$, which also controls every pair with the outside zero coordinate. Thus all five gradient values have diameter at most $1/x_a$.

For synergy, partition the source space into $E_1 \cap E_2$, $E_1 \setminus E_2$, $E_2 \setminus E_1$, and the outside region $O = (E_1 \cup E_2)^c$. For any law, let a, b, c, o be the masses of these four regions in that order. Define

$$\Phi(a, b, c) = \log \frac{(a+b)(a+c)}{a(a+b+c)}. \quad (62)$$

Its gradient values, including the outside region o , are

$$D_a = \frac{1}{a+b} + \frac{1}{a+c} - \frac{1}{a} - \frac{1}{a+b+c}, \quad (63)$$

$$D_b = \frac{1}{a+b} - \frac{1}{a+b+c}, \quad D_c = \frac{1}{a+c} - \frac{1}{a+b+c}, \quad D_o = 0. \quad (64)$$

Here $D_a \leq 0 \leq D_b, D_c$ and

$$D_b - D_a = \frac{1}{a} - \frac{1}{a+c}, \quad D_c - D_a = \frac{1}{a} - \frac{1}{a+b}. \quad (65)$$

Thus $\text{osc } D \leq 1/a$. The informative and misinformative synergy components are diamond functions on the full and target-restricted partitions.

For a path law $Q = p_s$, define

$$(x_a, x_b, x_c, x_o) = (Q(E_1 \cap E_2 \cap C), Q((E_1 \setminus E_2) \cap C), Q((E_2 \setminus E_1) \cap C), Q(O \cap C)), \quad (66)$$

and define (y_a, y_b, y_c, y_o) by replacing C with C^c . The keyed cell gives $x_a \geq \mu_s > 0$. Addition is componentwise, and

$$G_\Phi(x, y) = \Phi(x+y) - \Phi(x). \quad (67)$$

For $i \in \{a, b, c, o\}$, the y_i -gradient coordinate is $F_i = D_i(x+y)$ and the x_i -gradient coordinate is $X_i = F_i - D_i(x)$. These are the eight gradient values.

The six F - F pairs use the ordinary diamond bound. Among the six X - X pairs, $X_a - X_b$ and $X_a - X_c$ are differences of nonnegative diamond gaps in $[0, 1/x_a]$, and $X_b - X_c$ is a difference of reciprocal decrements in $[-1/x_a, 0]$. The ordinary diamond signs give $-D_a(v), D_b(v), D_c(v) \in [0, 1/v_a]$. Hence each F_i and X_i has magnitude at most $1/x_a$; this controls all pairs that contain $F_o = X_o = 0$. The four same-index pairs satisfy $F_i - X_i = D_i(x)$. The four crossed pairs between a and b or c are differences of nonnegative values in $[0, 1/x_a]$. Up to the symmetry $b \leftrightarrow c$, the only remaining crossed term is

$$F_b - X_c = H(x+y) + D_c(x), \quad H(v) = \frac{1}{v_a + v_b} - \frac{1}{v_a + v_c}. \quad (68)$$

If $H(x+y) \geq 0$, use $H(x+y) \leq 1/(x_a + x_b)$ and $D_c(x) - D_a(x) = 1/x_a - 1/(x_a + x_b)$ with $D_a(x) \leq 0$. If $H(x+y) < 0$, use $H(x+y) \geq -1/(x_a + x_c)$ and $0 \leq D_c(x) \leq 1/x_a$. The reverse inequality and the symmetric crossed term follow in the same way. Hence $\text{osc } \nabla G_\Phi \leq 1/x_a$. Lemma 6.1 proves Equation (54).

It remains to control the weight terms. For net unique information, let a, b be the target-region masses and let $a_f \geq a$, $b_f \geq b$ be the full-event masses. Then

$$\exp(\pi_{\text{unique}}^{\text{sx}}) = \frac{a(a_f + b_f)}{a_f(a+b)} \geq \frac{4a}{(1+a)^2} \geq \frac{4p_{\min}}{(1+p_{\min})^2}. \quad (69)$$

The first inequality uses $b_f \geq b$, $a_f + b_f \leq 1$, and $(1-b)(a+b) \leq (1+a)^2/4$. Under the p -law, this gives the $-J$ lower endpoint for each unique net atom. The upper endpoint is L because $N(a_f, b_f) \leq L$ and $N(a, b) \geq 0$. Moreover,

$$0 \leq \Phi(a, b, c) \leq J \quad \text{under the } p\text{-law.} \quad (70)$$

For the upper bound, put $u = a + b + c \geq a$. The arithmetic-geometric mean inequality gives $(a + b)(a + c) \leq (a + u)^2/4$. The ratio $(a + u)^2/(4au)$ is nondecreasing in $u \geq a$, so $u \leq 1$ reduces it to $(1 + a)^2/(4a)$. This last expression is nonincreasing on $(0, 1]$. Under the p -law, redundancy and unique net atoms lie in $[-J, L]$, and net synergy lies in $[-J, J]$. These ranges control the p -weighted term in the average-change split. The outer caps in the theorem table use support cardinality and therefore bound the averaged values under both laws. Centered total variation now gives the displayed atom-specific weight terms. This two-source result uses only the displayed log expressions and elementary range arguments. It does not use the published general-source component-nonnegativity result used in Theorem 6.3. $\square$

The lower endpoint in Equation (53) is attained as an event bound when $p_{\min} \leq 1/3$, the intersection has mass $p_{\min}$, and the two exclusive regions each have mass $(1 - p_{\min})/2$. A separate event construction attains the upper endpoint when $p_{\min} \leq 1/2$. These endpoint laws do not prove that a general Möbius-row bound is sharp.

The revision-1 constants $(\Lambda, 2\Lambda, 3\Lambda)$ for cumulative terms remain valid triangle-inequality baselines. They are superseded because they ignore the direct nested and intersection-PMI gradients. A generic two-point range extremizer retained in the executable record is not shown to be SxPID-realizable and proves no SxPID sharpness claim.

For a realizable two-source family, in lexicographic binary-cell order let

$$p_\varepsilon = \left(\varepsilon, \frac{1}{6}, \frac{1}{3}, \varepsilon, \frac{1}{2} - 5\varepsilon, \varepsilon, \varepsilon, \varepsilon \right), \quad 0 < \varepsilon < \frac{1}{12}. \quad (71)$$

Net synergy at 000 and 111 equals $-L + \log 10 + o(1)$ and $L - \log(40/3) + o(1)$. Their oscillation divided by L tends to two. Moving mass $0 < \rho < \varepsilon$ from 000 to 111 proves asymptotic necessity of the δL centered net-weight coefficient for that decomposition. For admissible $\rho < \varepsilon$, the Λ pointwise-value term can dominate the total bound. Thus this family does not prove global sharpness of the complete average bound.

The asymptotic statement is a prose derivation from the displayed family. The executable fixture checks only the finite case $\varepsilon = 1/60$ and its near-tight pointwise Λ ratios.

If $\delta \leq p_{\min}/2$, put $\xi = \delta/(2p_{\min}) \leq 1/4$. Lean also proves

$$\Lambda \leq \frac{\xi}{1 - \xi} \leq \frac{4\xi}{3} = \frac{2\delta}{3p_{\min}}. \quad (72)$$

No result in this section is a binary64 error theorem, a support-creation theorem, or a continuous-estimator theorem.

7 Composition with SxPID

For common-law sampling, empirical support is a subset of population support almost surely. On the event in Theorem 4.1, the strict condition

$$D_n < 2p_{\min} \quad (73)$$

forces empirical and population support to agree. The realized error satisfies $\delta \leq D_n$. The function $\Lambda(\delta, p_{\min})$ and every displayed right-hand side are nondecreasing in δ . Substitute D_n for δ in Equation (39), and use the result in Equations (44), (48), (49), (50), and (54), together with the two-source table. This gives simultaneous exact-real envelopes for every supported cumulative term, every fixed general-source Möbius atom, and every complete two-source atom.

A usable numerical envelope needs a justified lower bound on $p_{\min}$. The observed minimum frequency is not such a bound. A positive population cell can be absent from a finite sample.

For Theorem 5.1, also require every P_i to be supported inside $\text{supp}(P_*)$. Define

$$p_{\min, \star} = \min_{z \in \text{supp}(P_*)} P_*(z). \quad (74)$$

If

$$b_n + D_n < 2p_{\min, \star}, \quad (75)$$

then the realized error is at most $b_n + D_n$. Monotonicity gives the same composition with $b_n + D_n$ on the right-hand side.

A frozen finite-output transform can be handled conditionally on an independent training artifact. Conditional on that artifact, the color map must be fixed. Every row must have the stated conditional law. The complete rows in every color class must be mutually independent. A random conditional support floor gives a conditional result unless a deterministic lower bound is available.

The displayed common-law envelope proves almost-sure exact-real finite-alphabet SxPID plug-in consistency under the sufficient condition $V_n \log(n)/n^2 \rightarrow 0$; fixed d is sufficient. The displayed drift envelope proves almost-sure exact-real reference-law SxPID consistency when this sufficient condition holds and $b_n \rightarrow 0$. These are not necessary conditions under a stronger sampling theorem. The result does not validate continuous $I_{\cap}^{\text{sx}}$, arbitrary overlapping windows, adaptive preprocessing on evaluation rows, unknown mixing rates, or binary64 asymptotics.

8 Counterexamples and invalidated routes

Invalid route	Counterexample	Required correction
Pairwise independence inside one color	Let U be uniform on $\mathbb{F}_2^4$. Use the 15 nonzero linear forms $a \cdot U$ as binary rows. Every pair is independent. All rows are zero with probability $1/16$. The empirical L^1 error is then one, while the false one-color bound is $2e^{-15/2} \approx 0.00111$.	Require mutual independence of all complete rows in each color.
Remove the color factor	Take m independent fair bits and copy each bit once into each of d colors. Each color is i.i.d., but only $m = n/d$ independent values remain.	Keep V_n , or keep the coarser factor d .
Infer convergence from one color per row	Set every row equal to one common fair bit. Singleton colors satisfy their local premise, but $V_n = n^2$ and the empirical law does not converge to the fair law.	This coloring cannot prove convergence; use a stronger coloring or another theorem.
Read the envelope condition as necessary	Let the rows be i.i.d. but assign each row a separate color. Then $V_n = n^2$, although the empirical law converges by the strong law.	State $V_n \log(n)/n^2 \rightarrow 0$ only as a sufficient condition for this envelope.

Invalid route	Counterexample	Required correction
Data-adaptive coloring without the complete conditional contract	Set every row equal to one fair bit, then select one occupied color from that bit. Conditional rows are degenerate, but their conditional law is not the common unconditional law.	Require fixed colors, or prove the full conditional law and independence premises.
Substitute a mixing label for the color premise	Use a stationary fair two-state Markov chain with flip probability $\theta = 1/100$. Its transition dependence decays geometrically. For two rows, the empirical L^1 error is one with probability $1 - \theta = 0.99$, above the false one-color bound $2e^{-1}$.	Use a theorem with an explicit mixing coefficient and rate, or prove a valid independence coloring.
Pairwise lag independence as strong ℓ -dependence	The finite-field construction has pairwise independence and a joint failure.	Require the sigma-field definition. Residues modulo $\ell + 1$ then give valid colors by ordered induction.
Halve the net weight term from the generic range alone	On two points, let $w = (-\delta/2, \delta/2)$ and a generic row value be $f = (-r_\alpha L, r_\alpha L)$. Then $ \sum wf = r_\alpha L\delta$. This row value is not shown to be realizable by an SxPID atom when $r_\alpha > 1$. It proves no SxPID sharpness claim.	Keep the generic range baseline unless a separate SxPID argument supplies a smaller range.
Allow $\delta = 2p_{\min}$	Move all mass from a least-probable supported cell. Its required logs become undefined.	Keep $\delta < 2p_{\min}$ strict.
Control only separate marginals	Set $S_1 = S_2 = X$. For $0 < \rho < 1$, at the supported realization $(X, T) = (0, 0)$, use fair binary X, T marginals with diagonal cells $(1 + \rho)/4$ and off-diagonal cells $(1 - \rho)/4$, then replace ρ by $-\rho$. Marginals stay fixed, but pointwise redundancy changes by $\log((1 + \rho)/(1 - \rho))$.	Control the complete joint law.
Allow new support with a linear average bound	Let $p(0, 0, 0) = 1$, and let $q(0, 0, 0) = 1 - \varepsilon$, $q(1, 1, 1) = \varepsilon$. Then $\ q - p\ _1 = 2\varepsilon$, while the averaged redundancy is $-(1 - \varepsilon) \log(1 - \varepsilon) - \varepsilon \log \varepsilon$.	Require common support or use a nonlinear boundary modulus.
Use observed minimum frequency as a population floor	An unobserved positive cell has empirical frequency zero.	Use an external floor or a separate simultaneous lower-confidence result.

The executable challenge suite enumerates the finite-field pairwise-independence construction, copied-color construction, singleton-color construction, and adaptive-color construction. It also checks a support-deletion boundary, an unspecified-mixing construction, a generic net-weight range extremizer, a univariate-marginal-only construction, and a new-support construction. The remaining checks cover the telescoping allocation, class-size inequalities, high-precision one-log cases, and all displayed bounds on four committed two-source law pairs. One full-support pair has count vectors

$$(10, 100, 200, 10, 250, 10, 10, 10) \quad \text{and} \quad (1, 100, 200, 10, 250, 10, 10, 19). \quad (76)$$

It has $\delta = 3/100$, $\eta = 3/200$, $p_{\min} = 1/60$, and $p_{\min}/(p_{\min} - \eta) = 10$. Its largest observed synergy misinformative and net changes exceed 0.97Λ and 0.95Λ . This is bounded near-tightness evidence for the one- Λ pointwise constant, not a proof of global sharpness.

The fixed-window fixture uses independent fair innovations U_t, V_t and

$$S_{1,t} = U_t \text{ OR } U_{t+1}, \quad S_{2,t} = V_t \text{ OR } V_{t+1}, \quad T_t = S_{1,t} \text{ XOR } S_{2,t}. \quad (77)$$

Even and odd starts are the two colors. Its exact joint count weights are $1 : 3 : 3 : 9$. The high-precision Decimal oracle supplies expected logarithmic values. For stored logarithmic constants and bounds, the Rust reconstruction tolerance is $32\varepsilon_{\text{mach}} \max(1, |x_{\text{Rust}}|, |x_{\text{oracle}}|)$. For categorical estimator outputs, the absolute comparison ceiling is $32\varepsilon_{\text{mach}}$ nats, where $\varepsilon_{\text{mach}} = 2^{-52}$ for binary64. The test also checks that row reversal gives bit-identical output.

9 Formal and executable evidence boundary

The Lean project is pinned to one toolchain and one exact dependency manifest. The repository checker rejects admitted proof placeholders, builds the project, and replays its declarations with Lean’s kernel checker.

The checked Lean project proves the following results across the `Dependence.lean` and `LocalContinuity.lean` source files:

- the event-mass $\delta/2$ lemma and exact attainment by the positive-coordinate event;
- positivity under the strict support margin;
- the one-log modulus, its monotonicity, and its rational simplification;
- the centered zero-sum oscillation inequality;
- reciprocal-diameter bounds for negative-event-log, nested-log-ratio, and intersection-PMI gradient coordinates;
- the ordinary two-source diamond gradient diameter and the algebraic and logarithmic $0 \leq \Phi \leq J$ bounds;
- the five-coordinate conditioned nested-event gradient diameter;
- the effective-color proxy bounds and the normalized factor range;
- the generic absolute linear-row bound;
- component and net range transfer, centered weight bounds, and combined finite-average perturbation bounds;
- the factorized positive segment-event floor;
- the exact unit-scale finite telescoping allocation; and
- the algebraic radius-exponent cancellation.

Lean does not encode random variables or conditional laws. It does not encode generalized Hölder, Hoeffding’s lemma, the Chernoff argument, or the probability union bound. The drift result and fixed-window independence argument also remain outside Lean. Differentiation, path integration, and the identification of event masses with algebraic gradient coordinates remain outside Lean. The eight-coordinate conditioned-diamond result used for net synergy is

outside the checked formalization. Lean does not encode SxPID events, the redundancy lattice, the published pointwise component-nonnegativity theorem, or the identification of its generic lemmas with SxPID atoms. Rust refinement and binary64 rounding remain outside Lean. The probability theorem is a prose proof. Lean machine-checks only the deterministic obligations listed above. The Python suite uses exact rational arithmetic for finite probability and count identities, plus 100-digit Decimal arithmetic for logarithms. Transcendental logarithms are not exact rational values. From each local count-table pair, the Rust test independently reconstructs both empirical laws, δ , η , $p_{\min}$, Λ , L , h , J , the atom family, and every stored bound before it checks the implementation outputs. It uses the scale-aware reconstruction tolerance and the absolute categorical-output ceiling declared above. These checks are bounded executable evidence. They are not a general proof, a global binary64 bound, or external review.

10 Ecosystem use boundary

Consumer	Result supplied here	Still required
Prisoma	Categorical SxPID law control after a frozen finite map when held-out rows meet the color contract	Concrete row law, justified support floor, independent transform receipt, and a valid window coloring
Galadriel	Dependence-aware categorical alternative to an i.i.d. plug-in claim	Removal of same-window fitting and generic jitter routes; explicit observation and coloring contracts
Haldir	Mathematical target for block or residue-class designs	Direct dependency, corrected atom interpretation, and proof of strong ℓ -dependence or another valid coloring
Crebain	Finite-alphabet validation component for a future integration	Exact observation mapping, run-log contract, direct dependency, and explicit scientific estimand

This document does not assert that a consumer meets these premises. Integration remains `not_claimed` until the consumer records and checks them.

11 Reproduction

From the repository root, run

```
python3 scripts/check-lean-finite-convergence.py
python3 scripts/generate-dependency-colored-sxpid-oracle.py
cargo test --locked -p pid-core --test dependency_colored_sxpid_oracle
scripts/check-dependency-colored-sxpid-pdf.sh
```

The PDF checker rebuilds this source with a fixed source-date epoch, rejects LaTeX diagnostics, and compares the exact PDF bytes. The Lean checker binds the toolchain, manifest, source set, dependency revisions, dependency origins, and clean checkout state. It removes ambient Git routing and global or system configuration while retaining each dependency checkout's local origin configuration for verification.

12 Conclusion

The result supplies a dependence-aware finite-sample path from a declared finite-row coloring to exact-real categorical shared-exclusions PID bounds. Its displayed envelope proves almost-sure

exact-real consistency under the sufficient condition $V_n \log(n)/n^2 \rightarrow 0$. The corresponding reference-law result also uses $b_n \rightarrow 0$. These are not necessary conditions under a stronger sampling theorem. The result separates four objects that must remain separate: sampling fluctuation, deterministic drift, support margin, and program arithmetic. The color-size proxy improves the proof constant for unbalanced classes. The telescoping allocation gives a simple time-uniform statement. The local logarithmic modulus gives a direct law-to-atom transfer once common support is justified. Every weaker route listed in Section 8 is retained as invalidated work.

References

- [1] A. Makkeh, A. J. Gutknecht, and M. Wibral. Introducing a differentiable measure of pointwise shared information. *Physical Review E*, 103:032149, 2021. <https://doi.org/10.1103/PhysRevE.103.032149>.
- [2] A. J. Gutknecht, M. Wibral, and A. Makkeh. Bits and pieces: understanding information decomposition from part-whole relationships and formal logic. *Proceedings of the Royal Society A*, 477:20210110, 2021. <https://doi.org/10.1098/rspa.2021.0110>.
- [3] W. Hoeffding. Probability inequalities for sums of bounded random variables. *Journal of the American Statistical Association*, 58(301):13–30, 1963. <https://doi.org/10.1080/01621459.1963.10500830>.
- [4] S. Janson. Large deviations for sums of partly dependent random variables. *Random Structures & Algorithms*, 24(3):234–248, 2004. <https://doi.org/10.1002/rsa.20008>.
- [5] C. Pelekis, J. Ramon, and Y. Wang. Hölder-type inequalities and their applications to concentration and correlation bounds. *Indagationes Mathematicae*, 28(1):170–182, 2017. <https://doi.org/10.1016/j.indag.2016.11.017>.
- [6] T. Weissman, E. Ordentlich, G. Seroussi, S. Verdú, and M. J. Weinberger. Inequalities for the L^1 deviation of the empirical distribution. HPL-2003-97 (R.1), 2003. <https://shiftright.com/mirrors/www.hpl.hp.com/techreports/2003/HPL-2003-97R1.pdf>.